Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

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Worked Solutions **Triangles and Lines**

Q001 **Written**

Determine the shape of triangle ABC with A $(3,\ \ 2)$, B $(-1,\ \ 0)$, C $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q002 **Written**

Determine the shape of triangle ABC with A $(2,\ \ 2)$, B $(2,\ \ 0)$, C $\left(2+\sqrt{3},\ \ 1\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q003 *MCQ*

Determine the shape of triangle ABC with A $(3,\ \ 2)$, B $(-1,\ \ 0)$, C $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: D

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q004 MCQ

Determine the shape of triangle ABC with A $(2,\ \ 2)$, B $(2,\ \ 0)$, C $\left(2+\sqrt{3},\ \ 1\right)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $equilateral$.

Correct option: C

Final answer

$equilateral$

Q005 Written

Determine the shape of triangle ABC with A $(-1,\ \ 2)$, B $(1,\ \ 1)$, C $(0,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2+b^2=c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right-angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q006 Written

Determine the shape of triangle ABC with A $(0,\ \ 2)$, B $(-1,\ \ -2)$, C $(4,\ \ 1)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Q007 MCQ

Determine the shape of triangle ABC with A $(-1,\ \ 2)$, B $(1,\ \ 1)$, C $(0,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ A$.

Correct option: B

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q008 MCQ

Determine the shape of triangle ABC with A $(0,\ \ 2)$, B $(-1,\ \ -2)$, C $(4,\ \ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ A$

Q009 Written

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2+b^2=c^2$, taking c as the longest side.
3. Final classification: $scalene\ right-angled;\ right\ angle\ at\ A$.

Final answer

$scalene\ right-angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q010 MCQ

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right - angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$scalene\ right - angled;\ right\ angle\ at\ A$

Q011 Written

Determine the shape of triangle ABC with A $(4,\ \ 9)$, B $(0,\ \ 6)$, C $(3,\ \ 2)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q012 **Written**

Determine the shape of triangle ABC with A $(1,\ \ 2)$, B $(5,\ \ 8)$, C $(11,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Q013 **Written**

Determine the shape of triangle ABC with A $(-1,\ \ 1)$, B $(4,\ \ 2)$, C $(1,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ C$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q014 MCQ

Determine the shape of triangle ABC with A $(4,\ 9)$, B $(0,\ 6)$, C $(3,\ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ B$.

Correct option: B

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ B$

Q015 MCQ

Determine the shape of triangle ABC with A $(1,\ 2)$, B $(5,\ 8)$, C $(11,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ B$.

Correct option: C

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q016 MCQ

Determine the shape of triangle ABC with A $(-1,\ \ 1)$, B $(4,\ \ 2)$, C $(1,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ C$.

Correct option: B

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q017 Challenge

If A $(1,\ 2)$, B $(2k,\ k)$, C $(4,\ 5)$ and ABC is isosceles and right-angled at B, find k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $k = 2$.

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q018 Challenge

If A $(1,\ \ 2)$, B $(2k,\ \ k)$, C $(4,\ \ 5)$ and ABC is isosceles and right-angled at B, find k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $k = 2$.

Correct option: C

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -4$.

Final answer

$y = -4$

Q020 Written

Find the equation of the line through (-3,8) parallel to the y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Q022 Written

Find the equation of the line through (3,5) perpendicular to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q023 Written

Find the equation of the line through (-4,5) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 5$.

Final answer

$y = 5$

Q024 Written

Find the equation of the line through (2,1) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 1$.

Final answer

$y = 1$

Worked Solutions **Triangles and Lines**

Q025 **Written**

Find the equation of the line through (-4,-7) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -7$.

Final answer

$y = -7$

Q026 **Written**

Find the equation of the line through (1,-2) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 1$.

Final answer

$x = 1$

Worked Solutions **Triangles and Lines**

Q027 **Written**

Find the equation of the line through (-3,-4) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q028 **Written**

Find the equation of the line through (3,-5) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 2$.

Final answer

$x = 2$

Q030 Written

Find the equation of the line through (4,3) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Q031 Written

Find the equation of the line through (-6,-2) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -6$.

Final answer

$x = -6$

Q032 MCQ

Find the equation of the line through (3,-4) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -4$.

Correct option: B

Final answer

$y = -4$

Worked Solutions **Triangles and Lines**

Q033 MCQ

Find the equation of the line through (-3,8) parallel to the y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: C

Final answer

$x = -3$

Q034 MCQ

Find the equation of the line through (2,6) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: D

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q036 MCQ

Find the equation of the line through (-4,5) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 5$.

Correct option: B

Final answer

$y = 5$

Worked Solutions **Triangles and Lines**

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 1$.

Correct option: D

Final answer

$y = 1$

Q038 MCQ

Find the equation of the line through (-4,-7) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -7$.

Correct option: B

Final answer

$y = -7$

Worked Solutions **Triangles and Lines**

Q039 *MCQ*

Find the equation of the line through (1,-2) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 1$.

Correct option: C

Final answer

$x = 1$

Q040 *MCQ*

Find the equation of the line through (-3,-4) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: D

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q042 MCQ

Find the equation of the line through (2,-7) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 2$.

Correct option: B

Final answer

$x = 2$

Worked Solutions **Triangles and Lines**

Q043 *MCQ*

Find the equation of the line through (4,3) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: D

Final answer

$x = 4$

Q044 *MCQ*

Find the equation of the line through (-6,-2) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -6$.

Correct option: A

Final answer

$x = -6$

Worked Solutions **Triangles and Lines**

Q045 **Written**

Find the equation of the line through (-3,1) and (-3,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q046 **Written**

Find the equation of the line through (-2,3) and (-5,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q047 **Written**

Find the equation of the line through (1,-1) and (4,0).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x/3 - 4/3(\text{equivalently}x - 3y - 4 = 0)$.

Final answer

$y = x/3 - 4/3(\text{equivalently}x - 3y - 4 = 0)$

Q048 **Written**

Find the equation of the line through (-2,-2) and (2,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q049 Written

Find the equation of the line through (-4,2) and (-4,8).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -4$.

Final answer

$x = -4$

Q050 Written

Find the equation of the line through (-2,-6) and (-4,1).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $7x + 2y + 26 = 0$.

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q051 **Written**

Find the equation of the line through (4,0) and (4,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Q052 **Written**

Find the equation of the line through (-1,2) and (-1,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -1$.

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q053 Written

Find the equation of the line through (-5,-3) and (-5,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -5$.

Final answer

$x = -5$

Q054 Written

Find the equation of the line through (-2,3) and (-3,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q055 **Written**

Find the equation of the line through (4,-2) and (1,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Q056 **Written**

Find the equation of the line through (-3,6) and (2,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q057 Written

Find the equation of the line through (1,2) and (3,10).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 4x - 2$.

Final answer

$y = 4x - 2$

Q058 Written

Find the equation of the line through (-1,-8) and (4,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x - 6$.

Final answer

$y = 2x - 6$

Worked Solutions **Triangles and Lines**

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: B

Final answer

$x = -3$

Q060 MCQ

Find the equation of the line through (-2,3) and (-5,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: C

Final answer

$y = 3$

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$.

Correct option: D

Final answer

$y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$

Q062 MCQ

Find the equation of the line through (-2,-2) and (2,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: A

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q063 MCQ

Find the equation of the line through (-4,2) and (-4,8):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -4$.

Correct option: B

Final answer

$x = -4$

Q064 MCQ

Find the equation of the line through (-2,-6) and (-4,1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $7x + 2y + 26 = 0$.

Correct option: D

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: C

Final answer

$x = 4$

Q066 MCQ

Find the equation of the line through (-1,2) and (-1,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -1$.

Correct option: A

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q067 MCQ

Find the equation of the line through (-5,-3) and (-5,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -5$.

Correct option: C

Final answer

$x = -5$

Q068 MCQ

Find the equation of the line through (-2,3) and (-3,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: D

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: D

Final answer

$y = -2$

Q070 MCQ

Find the equation of the line through (-3,6) and (2,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: C

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 4x - 2$.

Correct option: D

Final answer

$y = 4x - 2$

Q072 MCQ

Find the equation of the line through (-1,-8) and (4,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x - 6$.

Correct option: D

Final answer

$y = 2x - 6$

Worked Solutions **Triangles and Lines**

Q073 Written

Find the equation of the line through (2,-1) with gradient 3.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3x - 7$.

Final answer

$y = 3x - 7$

Q074 Written

Find the equation of the line through (-3,4) with gradient 2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x + 10$.

Final answer

$y = 2x + 10$

Worked Solutions **Triangles and Lines**

Q075 Written

Find the equation of the line through (-1,5) with gradient -1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -x + 4$.

Final answer

$y = -x + 4$

Q076 Written

Find the equation of the line through (-3,-8) with gradient 1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x - 5$.

Final answer

$y = x - 5$

Worked Solutions **Triangles and Lines**

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2x + 3$.

Final answer

$y = -2x + 3$

Q078 MCQ

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: D

Final answer

$y = 3x - 7$

Worked Solutions **Triangles and Lines**

Q079 *MCQ*

Find the equation of the line through (-3,4) with gradient 2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x + 10$.

Correct option: C

Final answer

$y = 2x + 10$

Q080 *MCQ*

Find the equation of the line through (-1,5) with gradient -1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -x + 4$.

Correct option: D

Final answer

$y = -x + 4$

Worked Solutions **Triangles and Lines**

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x - 5$.

Correct option: B

Final answer

$y = x - 5$

Q082 MCQ

Find the equation of the line through (2,-1) with gradient -2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2x + 3$.

Correct option: D

Final answer

$y = -2x + 3$

Worked Solutions **Triangles and Lines**

Q083 **Challenge**

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $y = 2$.

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q084 **Challenge**

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2$.

Correct option: D

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q085 Written

A $(2,\ \ 3)$, B $(4,\ \ -1)$, and C $(a-1,\ \ 2-a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Q086 Written

A $(2,\ \ 5)$, B $(4,\ \ 9)$, and C $(-1,\ \ a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = -1$.

Final answer

$a = -1$

Worked Solutions **Triangles and Lines**

Q087 Written

A $(1, \ -1)$, B $(-2, \ -3)$, and C $(4, \ a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 1$.

Final answer

$a = 1$

Q088 Written

A $(1, \ 2)$, B $(3, \ 1)$, and C $(a, \ -1)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Worked Solutions **Triangles and Lines**

Q089 Written

A $(-3,\quad -9)$, B $(5,\quad 7)$, and C $(a+3,\quad 3a-2)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a=5$.

Final answer

$a=5$

Q090 MCQ

A $(2,\quad 3)$, B $(4,\quad -1)$, and C $(a-1,\quad 2-a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a=7$.

Correct option: D

Final answer

$a=7$

Worked Solutions **Triangles and Lines**

Q091 MCQ

A $(2, \quad 5)$, B $(4, \quad 9)$, and C $(-1, \quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1$.

Correct option: D

Final answer

$a = -1$

Q092 MCQ

A $(1, \quad -1)$, B $(-2, \quad -3)$, and C $(4, \quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$.

Correct option: C

Final answer

$a = 1$

Worked Solutions **Triangles and Lines**

Q093 MCQ

A $(1,\ \ 2)$, B $(3,\ \ 1)$, and C $(a,\ \ -1)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D

Final answer

$a = 7$

Q094 MCQ

A $(-3,\ \ -9)$, B $(5,\ \ 7)$, and C $(a+3,\ \ 3a-2)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 5$.

Correct option: D

Final answer

$a = 5$

Worked Solutions **Triangles and Lines**

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 2 = 0$.

Final answer

$x + y - 2 = 0$

Q096 Written

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + 3y - 23 = 0$.

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q097 **Written**

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 1 = 0$.

Final answer

$x + y - 1 = 0$

Q098 **Written**

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $2x - 5y + 1 = 0$.

Final answer

$2x - 5y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 2 = 0$.

Correct option: B

Final answer

$x + y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 23 = 0$.

Correct option: A

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: B

Final answer

$x + y - 1 = 0$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 5y + 1 = 0$.

Correct option: D

Final answer

$2x - 5y + 1 = 0$

Worked Solutions

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $3x + 13y = 0$.

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q104 **Challenge**

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 13y = 0$.

Correct option: C

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q105 **Written**

Find the gradient of line $y = 2x + 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Q106 **Written**

Find the gradient of line $y = -3x + 2$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q107 Written

Find the gradient of line $x - 3y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1/3$.

Final answer

$m = 1/3$

Q108 Written

Find the gradient of line $2x - y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions   **Triangles and Lines**

Q109 **Written**

Find the gradient of line $x + 3y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q110 **Written**

Find the gradient of line $y = 3x + 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q111 Written

Find the gradient of line $y = 4x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 4$.

Final answer

$m = 4$

Q112 Written

Find the gradient of line $y = -4x - 7$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q113 Written

Find the gradient of line $6x - 2y = 5$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Q114 Written

Find the gradient of line $8x + 2y - 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q115 **Written**

Find the gradient of line $x + 4y - 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/4$.

Final answer

$m = -1/4$

Q116 **Written**

Find the gradient of line $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q117 Written

Find the gradient of line $y = -3x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q118 Written

Find the gradient of line $y = x - 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q119 **Written**

Find the gradient of line $6x + 2y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q120 **Written**

Find the gradient of line $2x - 2y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q121 **Written**

Find the gradient of line $x + 3y - 2 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q122 **Written**

Find the gradient of line $y = 2x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q123 **Written**

Find the gradient of line $y = -x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q124 **Written**

Find the gradient of line $y = 3x/2 + 6$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3/2$.

Final answer

$m = 3/2$

Worked Solutions **Triangles and Lines**

Q125 **Written**

Find the gradient of line $x + y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q126 **Written**

Find the gradient of line $2x + 3y + 8 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -2/3$.

Final answer

$m = -2/3$

Worked Solutions **Triangles and Lines**

Q127 Written

Find the gradient of line $3x + 6y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/2$.

Final answer

$m = -1/2$

Q128 MCQ

Find the gradient of line $y = 2x + 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q129 MCQ

Find the gradient of line $y = -3x + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Q130 MCQ

Find the gradient of line $x - 3y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1/3$.

Correct option: A

Final answer

$m = 1/3$

Worked Solutions Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: B

Final answer

$m = 2$

Q132 MCQ

Find the gradient of line $x + 3y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q133 MCQ

Find the gradient of line $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 4$.

Correct option: B

Final answer

$m = 4$

Worked Solutions **Triangles and Lines**

Q135 MCQ

Find the gradient of line $y = -4x - 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: D

Final answer

$m = -4$

Q136 MCQ

Find the gradient of line $6x - 2y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: C

Final answer

$m = -4$

Q138 MCQ

Find the gradient of line $x + 4y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/4$.

Correct option: C

Final answer

$m = -1/4$

Worked Solutions **Triangles and Lines**

Q139 MCQ

Find the gradient of line $y = 3x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: D

Final answer

$m = 3$

Q140 MCQ

Find the gradient of line $y = -3x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: A

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q141 MCQ

Find the gradient of line $y = x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: D

Final answer

$m = 1$

Q142 MCQ

Find the gradient of line $6x + 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: C

Final answer

$m = 1$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q145 MCQ

Find the gradient of line $y = 2x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Q146 MCQ

Find the gradient of line $y = -x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: A

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3/2$.

Correct option: B

Final answer

$m = 3/2$

Q148 MCQ

Find the gradient of line $x + y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: D

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -2/3$.

Correct option: B

Final answer

$m = -2/3$

Q150 MCQ

Find the gradient of line $3x + 6y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/2$.

Correct option: B

Final answer

$m = -1/2$

Worked Solutions **Triangles and Lines**

Q151 Written

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 14 = 0$.

Final answer

$3x - 2y - 14 = 0$

Q152 Written

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + y + 1 = 0$.

Final answer

$2x + y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q153 **Written**

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x - y - 4 = 0$.

Final answer

$2x - y - 4 = 0$

Q154 **Written**

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y - 4 = 0$.

Final answer

$2x + 3y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q155 Written

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x + 2y - 7 = 0$.

Final answer

$3x + 2y - 7 = 0$

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 14 = 0$.

Correct option: C

Final answer

$3x - 2y - 14 = 0$

Worked Solutions **Triangles and Lines**

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + y + 1 = 0$.

Correct option: D

Final answer

$2x + y + 1 = 0$

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y - 4 = 0$.

Correct option: C

Final answer

$2x - y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y - 4 = 0$.

Correct option: D

Final answer

$2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 2y - 7 = 0$.

Correct option: D

Final answer

$3x + 2y - 7 = 0$

Worked Solutions **Triangles and Lines**

Q161 **Written**

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $5x - 3y + 25 = 0$.

Final answer

$5x - 3y + 25 = 0$

Q162 **Written**

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 4 = 0$.

Final answer

$3x - 2y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q163 **Written**

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $x - 2y - 12 = 0$.

Final answer

$x - 2y - 12 = 0$

Q164 **Written**

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 5y - 13 = 0$.

Final answer

$2x + 5y - 13 = 0$

Worked Solutions **Triangles and Lines**

Q165 Written

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y + 19 = 0$.

Final answer

$2x + 3y + 19 = 0$

Q166 MCQ

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $5x - 3y + 25 = 0$.

Correct option: D

Final answer

$5x - 3y + 25 = 0$

Worked Solutions **Triangles and Lines**

Q167 *MCQ*

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 4 = 0$.

Correct option: D

Final answer

$3x - 2y - 4 = 0$

Q168 *MCQ*

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x - 2y - 12 = 0$.

Correct option: B

Final answer

$x - 2y - 12 = 0$

Worked Solutions **Triangles and Lines**

Q169 MCQ

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 5y - 13 = 0$.

Correct option: A

Final answer

$2x + 5y - 13 = 0$

Q170 MCQ

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y + 19 = 0$.

Correct option: D

Final answer

$2x + 3y + 19 = 0$

Worked Solutions **Triangles and Lines**

Q171 **Challenge**

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $m/k = 2\,(k = 3,\quad m = 6)$.

Final answer

$m/k = 2\,(k = 3,\quad m = 6)$

Worked Solutions **Triangles and Lines**

Q172 **Challenge**

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m/k = 2\,(k = 3, \quad m = 6)$.

Correct option: D

Final answer

$m/k = 2\,(k = 3, \quad m = 6)$

Worked Solutions **Triangles and Lines**

Q173 **Written**

For A $(5,\quad 11)$, B $(1,\quad -1)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $x + 3y - 18 = 0$.

Final answer

$x + 3y - 18 = 0$

Q174 **Written**

For A $(-1,\quad 2)$, B $(3,\quad -4)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - 3y - 5 = 0$.

Final answer

$2x - 3y - 5 = 0$

Worked Solutions **Triangles and Lines**

Q175 Written

Find the perpendicular bisector of segment A $(-2, \;\; 3)$, B $(2, \;\; 1)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 2 = 0$.

Final answer

$2x - y + 2 = 0$

Q176 Written

Find the perpendicular bisector of segment A $(-1, \;\; 2)$, B $(5, \;\; -2)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $3x - 2y - 6 = 0$.

Final answer

$3x - 2y - 6 = 0$

Worked Solutions **Triangles and Lines**

Q177 Written

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 1 = 0$.

Final answer

$2x - y + 1 = 0$

Q178 MCQ

For A $(5,\ \ 11)$, B $(1,\ \ -1)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 18 = 0$.

Correct option: A

Final answer

$x + 3y - 18 = 0$

Worked Solutions **Triangles and Lines**

Q179 MCQ

For A $(-1,\ 2)$, B $(3,\ -4)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 3y - 5 = 0$.

Correct option: C

Final answer

$2x - 3y - 5 = 0$

Q180 MCQ

Find the perpendicular bisector of segment A $(-2,\ 3)$, B $(2,\ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 2 = 0$.

Correct option: C

Final answer

$2x - y + 2 = 0$

Worked Solutions **Triangles and Lines**

Q181 *MCQ*

Find the perpendicular bisector of segment A $(-1,\ \ 2)$, B $(5,\ \ -2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 6 = 0$.

Correct option: D

Final answer

$3x - 2y - 6 = 0$

Q182 *MCQ*

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 1 = 0$.

Correct option: A

Final answer

$2x - y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q183 **Written**

Find the point B symmetric to A $(-4,\ \ 3)$ with respect to $3x - y + 5 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (2,\ \ 1)$.

Final answer

$B = (2,\ \ 1)$

Q184 **Written**

Find the point B symmetric to A $(0,\ \ 4)$ with respect to $2x - y - 1 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (4,\ \ 2)$.

Final answer

$B = (4,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q185 **Written**

Find the point B symmetric to A $(1,\ \ 2)$ with respect to $x + 2y - 10 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (3,\ \ 6)$.

Final answer

$B = (3,\ \ 6)$

Q186 **Written**

Find the point B symmetric to A $(4,\ \ -1)$ with respect to $4x - 2y - 3 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (-2,\ \ 2)$.

Final answer

$B = (-2,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q187 MCQ

Find the point B symmetric to A $(-4,\ 3)$ with respect to $3x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (2,\ 1)$.

Correct option: A

Final answer

$B = (2,\ 1)$

Q188 MCQ

Find the point B symmetric to A $(0,\ 4)$ with respect to $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (4,\ 2)$.

Correct option: A

Final answer

$B = (4,\ 2)$

Worked Solutions **Triangles and Lines**

Q189 *MCQ*

Find the point B symmetric to A $(1,\ \ 2)$ with respect to $x + 2y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (3,\ \ 6)$.

Correct option: C

Final answer

$B = (3,\ \ 6)$

Q190 *MCQ*

Find the point B symmetric to A $(4,\ \ -1)$ with respect to $4x - 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-2,\ \ 2)$.

Correct option: C

Final answer

$B = (-2,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q191 Challenge

Reflect A $(5,\quad 0)$ across the targeting line $y = 2x$. Find the receiver point B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $B = (-3,\quad 4)$.

Final answer

$B = (-3,\quad 4)$

Worked Solutions **Triangles and Lines**

Q192 **Challenge**

Reflect A $(5,\ \ 0)$ across the targeting line $y = 2x$. Find the receiver point B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-3,\ \ 4)$.

Correct option: C

Final answer

$B = (-3,\ \ 4)$

Worked Solutions **Triangles and Lines**

Q193 **Written**

Find the distance from (-2,1) to $3x - 2y + 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $3/\sqrt{13} = 3\sqrt{13}/13$.

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Q194 **Written**

Find the distance from (3,-2) to $x + 2y - 4 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{5}$.

Final answer

$\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q195 Written

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $6/\sqrt{13} = 6\sqrt{13}/13$.

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Q196 Written

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 6.

Final answer

6

Worked Solutions **Triangles and Lines**

Q197 Written

Find the perpendicular distance from (-8,1) to $y = 2x - 3$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $4\sqrt{5}$.

Final answer

$4\sqrt{5}$

Q198 Written

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 1.

Final answer

1

Worked Solutions **Triangles and Lines**

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{13}$.

Final answer

$\sqrt{13}$

Q200 Written

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 10.

Final answer

10

Worked Solutions **Triangles and Lines**

Q201 Written

Find the perpendicular distance from (1,2) to $y = 3x + 1$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/5$.

Final answer

$\sqrt{10}/5$

Q202 MCQ

Find the distance from (-2,1) to $3x - 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3/\sqrt{13} = 3\sqrt{13}/13$.

Correct option: A

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{5}$.

Correct option: D

Final answer

$\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $6/\sqrt{13} = 6\sqrt{13}/13$.

Correct option: D

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q205 MCQ

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 6.

Correct option: B

Final answer

6

Q206 MCQ

Find the perpendicular distance from (-8,1) to $y = 2x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $4\sqrt{5}$.

Correct option: D

Final answer

$4\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1.

Correct option: D

Final answer

1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{13}$.

Correct option: A

Final answer

$\sqrt{13}$

Worked Solutions **Triangles and Lines**

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: B

Final answer

10

Q210 MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/5$.

Correct option: D

Final answer

$\sqrt{10}/5$

Worked Solutions **Triangles and Lines**

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/2$.

Final answer

$\sqrt{10}/2$

Q212 Written

Find the perpendicular length from (-1,5) to $y = 3x - 2$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/2$.

Correct option: D

Final answer
$\sqrt{10}/2$

Q214 MCQ

Find the perpendicular length from (-1,5) to $y = 3x - 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: B

Final answer
$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q215 Challenge

Find the shortest distance from P $(-1, \quad 5)$ to the radiation line $y = x/3 + 2$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1,\ \ 5)$ to the radiation line $y = x/3 + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Worked Solutions Triangles and Lines

Q217 Written

Find the area of triangle ABC with A $(1,\ \ 1)$, B $(3,\ \ 7)$, C $(-3,\ \ -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 10.

Final answer

10

Q218 Written

Find the area for A $(1,\ \ 5)$, B $(-3,\ \ -3)$, C $(3,\ \ -6)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 30.

Final answer

30

Worked Solutions **Triangles and Lines**

Q219 Written

Find the area for A $(-2,\ \ 1)$, B $(2,\ \ -1)$, C $(0,\ \ 4)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 8.

Final answer

8

Q220 Written

Find the area for A $(0,\ \ -6)$, B $(2,\ \ -2)$, C $(1,\ \ 5)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 9.

Final answer

9

Worked Solutions **Triangles and Lines**

Q221 **Written**

Find the area for A $(-2, \quad 4)$, B $(-3, \quad -5)$, C $(5, \quad -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 34.

Final answer

34

Q222 **Written**

Find the area for A $(2, \quad 5)$, B $(0, \quad 1)$, C $(4, \quad 2)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 7.

Final answer

7

Worked Solutions **Triangles and Lines**

Q223 MCQ

Find the area of triangle ABC with A $(1,\ \ 1)$, B $(3,\ \ 7)$, C $(-3,\ \ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: D

Final answer

10

Q224 MCQ

Find the area for A $(1,\ \ 5)$, B $(-3,\ \ -3)$, C $(3,\ \ -6)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 30.

Correct option: C

Final answer

30

Worked Solutions **Triangles and Lines**

Q225 MCQ

Find the area for A $(-2,\quad 1)$, B $(2,\quad -1)$, C $(0,\quad 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 8.

Correct option: A

Final answer

8

Q226 MCQ

Find the area for A $(0,\quad -6)$, B $(2,\quad -2)$, C $(1,\quad 5)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 9.

Correct option: B

Final answer

9

Worked Solutions **Triangles and Lines**

Q227 MCQ

Find the area for A $(-2, \ 4)$, B $(-3, \ -5)$, C $(5, \ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 34.

Correct option: B

Final answer

34

Q228 MCQ

Find the area for A $(2, \ 5)$, B $(0, \ 1)$, C $(4, \ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 7.

Correct option: D

Final answer

7

Worked Solutions **Triangles and Lines**

Q229 **Challenge**

Find the area of the wound triangle A $(-2,\ \ 1)$, B $(1,\ \ 5)$, C $(-21,\ \ -21)$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 5.

Final answer

5

Worked Solutions

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2,\ \ 1)$, B $(1,\ \ 5)$, C $(-21,\ \ -21)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 5.

Correct option: D

Final answer

5

Worked Solutions **Triangles and Lines**

Q231 **Written**

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1/2, \ -3/2)$.

Final answer

$(1/2, \ -3/2)$

Q232 **Written**

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-2, \ 3)$.

Final answer

$(-2, \ 3)$

Worked Solutions **Triangles and Lines**

Q233 Written

The line $(k+2)x + ky - 6 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(3,\quad -3)$.

Final answer

$(3,\quad -3)$

Q234 Written

The line $(k+3)x - (2k+1)y + k - 2 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1,\quad 1)$.

Final answer

$(1,\quad 1)$

Worked Solutions **Triangles and Lines**

Q235 Written

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-3,\ \ 1)$.

Final answer

$(-3,\ \ 1)$

Q236 MCQ

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1/2,\ \ -3/2)$.

Correct option: D

Final answer

$(1/2,\ \ -3/2)$

Worked Solutions **Triangles and Lines**

Q237 MCQ

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-2,\quad 3)$.

Correct option: B

Final answer

$(-2,\quad 3)$

Q238 MCQ

The line $(k+2)x+ky-6=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3,\quad -3)$.

Correct option: D

Final answer

$(3,\quad -3)$

Worked Solutions **Triangles and Lines**

Q239 MCQ

The line $(k+3)x-(2k+1)y+k-2=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1,\ \ 1)$.

Correct option: D

Final answer

$(1,\ \ 1)$

Q240 MCQ

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-3,\ \ 1)$.

Correct option: A

Final answer

$(-3,\ \ 1)$

Worked Solutions **Triangles and Lines**

Q241 Challenge

The line $-(k-2)x+(k-3)y=-3k+5$ passes through a fixed point for every k. Find it.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $(4,\quad 1)$.

Final answer

$(4,\quad 1)$

Worked Solutions **Triangles and Lines**

Q242 **Challenge**

The line $-(k-2)x+(k-3)y=-3k+5$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(4, \quad 1)$.

Correct option: A

Final answer

$(4, \quad 1)$

Worked Solutions **Triangles and Lines**

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -1,\ -1/2,\ or 1$.

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions **Triangles and Lines**

Q245 Written

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 11/10,\ 3/2,\ or - 9/2$.

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 1,\ -3/2,\ or 2$.

Final answer

$a = 1,\ -3/2,\ or 2$

Worked Solutions **Triangles and Lines**

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -4,\ -3,\ or 6$.

Correct option: A

Final answer

$a = -4,\ -3,\ or 6$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1,\ -1/2,\ or 1$.

Correct option: A

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions **Triangles and Lines**

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 11/10,\ 3/2,\ or - 9/2$.

Correct option: C

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1,\ -3/2,\ or 2$.

Correct option: D

Final answer

$a = 1,\ -3/2,\ or 2$

Worked Solutions **Triangles and Lines**

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 15.

Final answer

15

Worked Solutions **Triangles and Lines**

Q252 **Challenge**

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 15.

Correct option: A

Final answer

15

Worked Solutions **Triangles and Lines**

Quiz MCQ 1 Concept Quiz · MCQ

Determine the shape of triangle A $(0,\ 0)$, B $(3,\ 0)$, C $(0,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right-angled$.

Correct option: C

Final answer

$scalene\ right-angled$

Quiz MCQ 2 Concept Quiz · MCQ

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: B

Final answer

$y = 3x - 7$

Worked Solutions **Triangles and Lines**

Quiz MCQ 3 Concept Quiz · MCQ

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $1/5$.

Correct option: D

Final answer

$1/5$

Quiz MCQ 4 Concept Quiz · MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3, \quad -3)$.

Correct option: C

Final answer

$(3, \quad -3)$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area of triangle A $(0,\ 0)$, B $(4,\ 0)$, C $(0,\ 3)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 6.

Final answer

6

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the perpendicular bisector of the segment joining A $(1,\ 2)$ and B $(5,\ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $y = -2x + 9$.

Final answer

$y = -2x + 9$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $y = -x/3 + 7/3$.

Final answer

$y = -x/3 + 7/3$

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Answer key

Use the question number shown on each page.

Q001 *equilateral*

Q002 *equilateral*

Q003 D

Q004 C

Q005 $isosceles\ right-angled;\ right\ angle\ at\ A$

Q006 $isosceles\ right-angled;\ right\ angle\ at\ A$

Q007 B

Q008 D

Q009 $scalene\ right-angled;\ right\ angle\ at\ A$

Q010 D

Q011 $isosceles\ right-angled;\ right\ angle\ at\ B$

Q012 $isosceles\ right-angled;\ right\ angle\ at\ B$

Q013 $isosceles\ right-angled;\ right\ angle\ at\ C$

Q014 B

Q015 C

Q016 B

Q017 $k = 2$

Q018 C

Q019 $y = -4$

Q020 $x = -3$

Q021 $y = 6$

Q022 $x = 3$

Q023 $y = 5$

Q024 $y = 1$

Q025 $y = -7$

Q026 $x = 1$

Q027 $x = -3$

Q028 $x = 3$

Q029 $x = 2$

Q030 $x = 4$

Q031 $x = -6$

Q032 B

Q033 C

Q034 D

Q035 D

Q036 B

Q037 D

Q038 B

Q039 C

Q040 D

Q041 D

Q042 B

Q043 D

Q044 A

Q045 $x = -3$

Q046 $y = 3$

Q047 $y = x/3 - 4/3(\text{equivalently}x - 3y - 4 = 0)$

Q048 $y = -2$

Q049 $x = -4$

Q050 $7x + 2y + 26 = 0$

Q051 $x = 4$

Q052 $x = -1$

Q053 $x = -5$

Q054 $y = 3$

Q055 $y = -2$

Q056 $y = 6$

Q057 $y = 4x - 2$

Q058 $y = 2x - 6$

Q059 B

Q060 C

Q061 D

Q062 A

Q063 B

Q064 D

Q065 C

Q066 A

Q067 C

Q068 D

Q069 D

Q070 C

Q071 D

Q072 D

Q073 $y = 3x - 7$

Q074 $y = 2x + 10$

Q075 $y = -x + 4$

Q076 $y = x - 5$

Q077 $y = -2x + 3$

Q078 D

Q079 C

Q080 D

Q081 B

Q082 D

Q083 $y = 2$

Q084 D

Q085 $a = 7$

Q086 $a = -1$

Q087 $a = 1$

Q088 $a = 7$

Q089 $a = 5$

Q090 D

Q091 D

Q092 C

Q093 D

Q094 D

Q095 $x + y - 2 = 0$

Q096 $x + 3y - 23 = 0$

Q097 $x + y - 1 = 0$

Q098 $2x - 5y + 1 = 0$

Q099 B

Q100 A

Q101 B

Q102 D

Q103 $3x + 13y = 0$

Q104 C

Q105 $m = 2$
Q106 $m = -3$
Q107 $m = 1/3$
Q108 $m = 2$
Q109 $m = -1/3$
Q110 $m = 3$
Q111 $m = 4$
Q112 $m = -4$
Q113 $m = 3$
Q114 $m = -4$
Q115 $m = -1/4$
Q116 $m = 3$
Q117 $m = -3$
Q118 $m = 1$
Q119 $m = -3$
Q120 $m = 1$
Q121 $m = -1/3$
Q122 $m = 2$
Q123 $m = -1$
Q124 $m = 3/2$
Q125 $m = -1$
Q126 $m = -2/3$
Q127 $m = -1/2$
Q128 D
Q129 C
Q130 A
Q131 B
Q132 A
Q133 A
Q134 B
Q135 D
Q136 A
Q137 C
Q138 C
Q139 D
Q140 A
Q141 D
Q142 C
Q143 C
Q144 A
Q145 D
Q146 A
Q147 B
Q148 D
Q149 B
Q150 B
Q151 $3x - 2y - 14 = 0$
Q152 $2x + y + 1 = 0$
Q153 $2x - y - 4 = 0$
Q154 $2x + 3y - 4 = 0$
Q155 $3x + 2y - 7 = 0$
Q156 C
Q157 D
Q158 C
Q159 D
Q160 D
Q161 $5x - 3y + 25 = 0$
Q162 $3x - 2y - 4 = 0$
Q163 $x - 2y - 12 = 0$
Q164 $2x + 5y - 13 = 0$
Q165 $2x + 3y + 19 = 0$
Q166 D
Q167 D
Q168 B
Q169 A
Q170 D
Q171 $m/k = 2\,(k = 3,\ \ m = 6)$
Q172 D
Q173 $x + 3y - 18 = 0$
Q174 $2x - 3y - 5 = 0$
Q175 $2x - y + 2 = 0$
Q176 $3x - 2y - 6 = 0$
Q177 $2x - y + 1 = 0$
Q178 A
Q179 C
Q180 C
Q181 D
Q182 A
Q183 $B = (2,\ \ 1)$
Q184 $B = (4,\ \ 2)$
Q185 $B = (3,\ \ 6)$
Q186 $B = (-2,\ \ 2)$
Q187 A
Q188 A
Q189 C
Q190 C
Q191 $B = (-3,\ \ 4)$
Q192 C
Q193 $3/\sqrt{13} = 3\sqrt{13}/13$
Q194 $\sqrt{5}$
Q195 $6/\sqrt{13} = 6\sqrt{13}/13$
Q196 6
Q197 $4\sqrt{5}$
Q198 1
Q199 $\sqrt{13}$
Q200 10
Q201 $\sqrt{10}/5$
Q202 A
Q203 D
Q204 D
Q205 B
Q206 D
Q207 D
Q208 A
Q209 B
Q210 D
Q211 $\sqrt{10}/2$
Q212 $\sqrt{10}$
Q213 D
Q214 B
Q215 $\sqrt{10}$
Q216 A
Q217 10
Q218 30
Q219 8
Q220 9
Q221 34
Q222 7
Q223 D
Q224 C
Q225 A
Q226 B
Q227 B
Q228 D
Q229 5
Q230 D
Q231 $(1/2,\ \ -3/2)$
Q232 $(-2,\ \ 3)$

Q233 $(3, \ -3)$

Q234 $(1, \ 1)$

Q235 $(-3, \ 1)$

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 $(4, \ 1)$

Q242 A

Q243 $a = -4, \ -3, \ or 6$

Q244 $a = -1, \ -1/2, \ or 1$

Q245 $a = 11/10, \ 3/2, \ or - 9/2$

Q246 $a = 1, \ -3/2, \ or 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$

Quiz WRITTEN 7 $y = -x/3 + 7/3$

Quiz WRITTEN 8 $a = -4, \ -3, \ or 6$